

Clustering-Based Pairs Trading

Replicating & extending Rotondi & Russo (2025)

Matched the paper's best strategy at Sharpe 1.028 (paper: 1.01)

21-year, survivorship-bias-free, out-of-sample backtest — built from the data up

Picking the right pairs is the hard part

- Pairs trading: trade two co-moving securities' spread — short the rich one, long the cheap one, profit when it reverts.
- Across a ~1,000-stock universe there are ~500,000 candidate pairs. Brute-forcing them is noisy and overfits badly.
- We want pairs that are economically related, not coincidental.

The paper's idea

Use unsupervised ML clustering to group similar stocks first, then only form pairs WITHIN a cluster — fewer, higher-quality candidates. My goal: rebuild it independently, reproduce the result, then extend it.

Built for honesty, not optics

Source	WRDS / CRSP daily — institutional-grade academic data
Period	Jan 2000 – Dec 2023 (24 years)
Universe	991 stocks — share codes 10/11 (US ordinary common only)
Survivorship bias	Eliminated — delisted names kept, delisting returns modelled
Trading window	2003–2023 = 251 monthly OUT-OF-SAMPLE returns

First 3 years are reserved purely as the formation window — so every reported return is genuinely out-of-sample. The 79 names dropped vs the raw pull are fully accounted for (foreign / REITs / funds).

METHOD

The pipeline — rolling, every month



Each month: form pairs on the trailing 3 years, trade them for the next 1 month, roll forward — 251 times.

Pairs are only ever formed between two stocks in the same cluster.

Two distance metrics

SSD

baseline · my Phase 1

Sum of Squared Deviations between normalised price paths. The classic Gatev et al. selector.

PC

the paper's winner

Partial-correlation distance on market-adjusted returns. Strips out the common market factor → finds idiosyncratic, market-neutral co-movement.

The distance feeds the clusterer — and this single choice is the biggest driver of risk-adjusted performance.

Clustering with OPTICS

- Density-based (cousin of DBSCAN) — no need to pre-specify the number of clusters.
- Leaves genuinely unrelated stocks UNclustered, instead of forcing them into a group.

Validation vs ground-truth industry codes (Dec 2023):

Metric	Mine SSD	Mine PC	Paper
# clusters	47	81	48 / 109
Purity vs SIC codes	0.871	0.937	0.81 / 0.84
GOOG & GOOGL together?	yes	yes	yes

xi tuned on cluster quality across 3 hold-out dates — and FROZEN before ever looking at Sharpe (no overfitting).

Cointegration filter & the trading signal

Engle-Granger filter (optional variant)

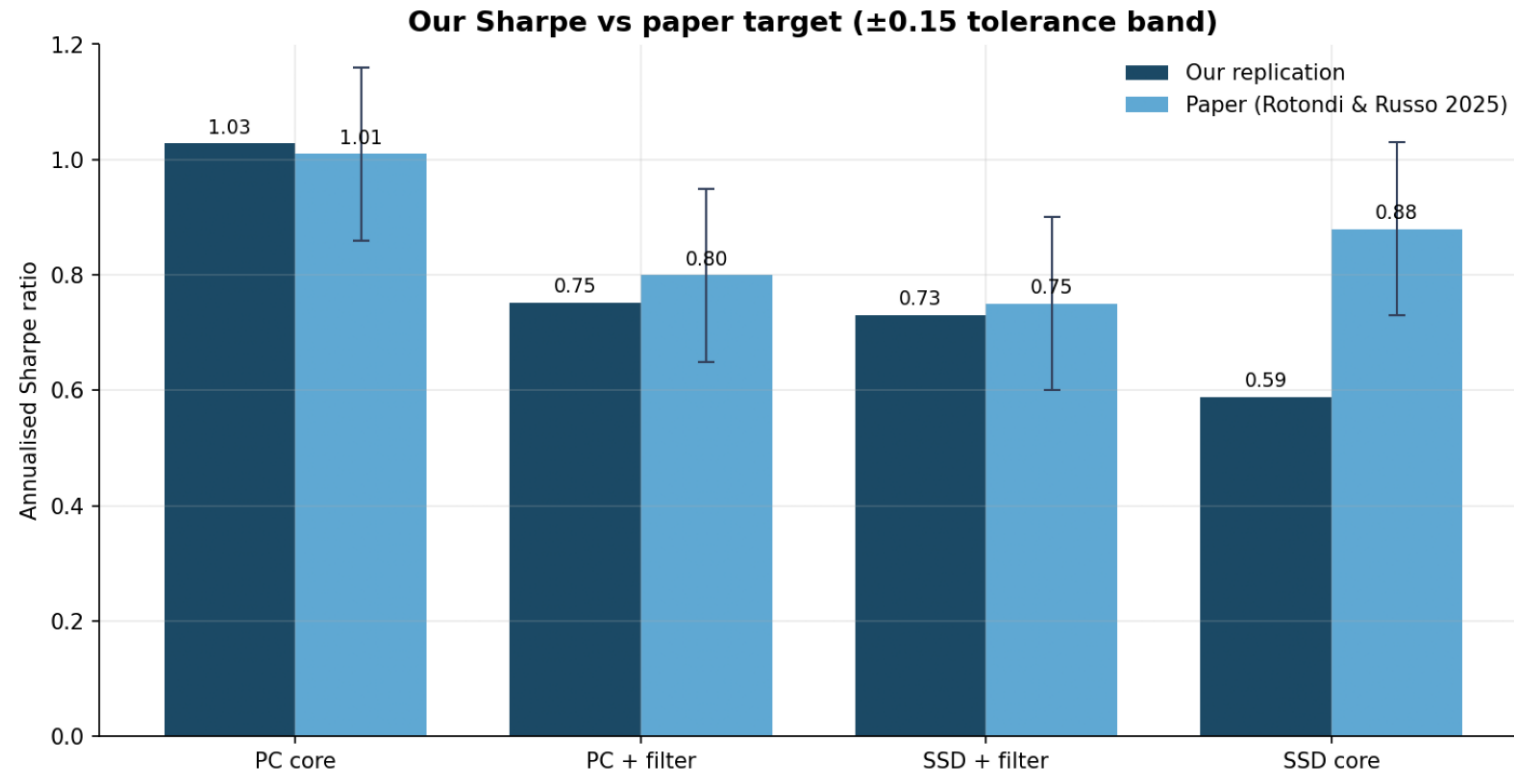
- Keep a pair only if the spread rejects the unit-root null at 5%, and
- spread half-life $\in [5, 60]$ trading days — fast enough to revert in the window, slow enough to not be noise.

Trading signal

- Hedge ratio γ via OLS $\rightarrow$ spread series $\rightarrow$ 6-month rolling z-score (strict look-ahead protection).
- Enter when $|z| > 2$ · exit at $z = 0$ · force-close at month-end.

RESULTS

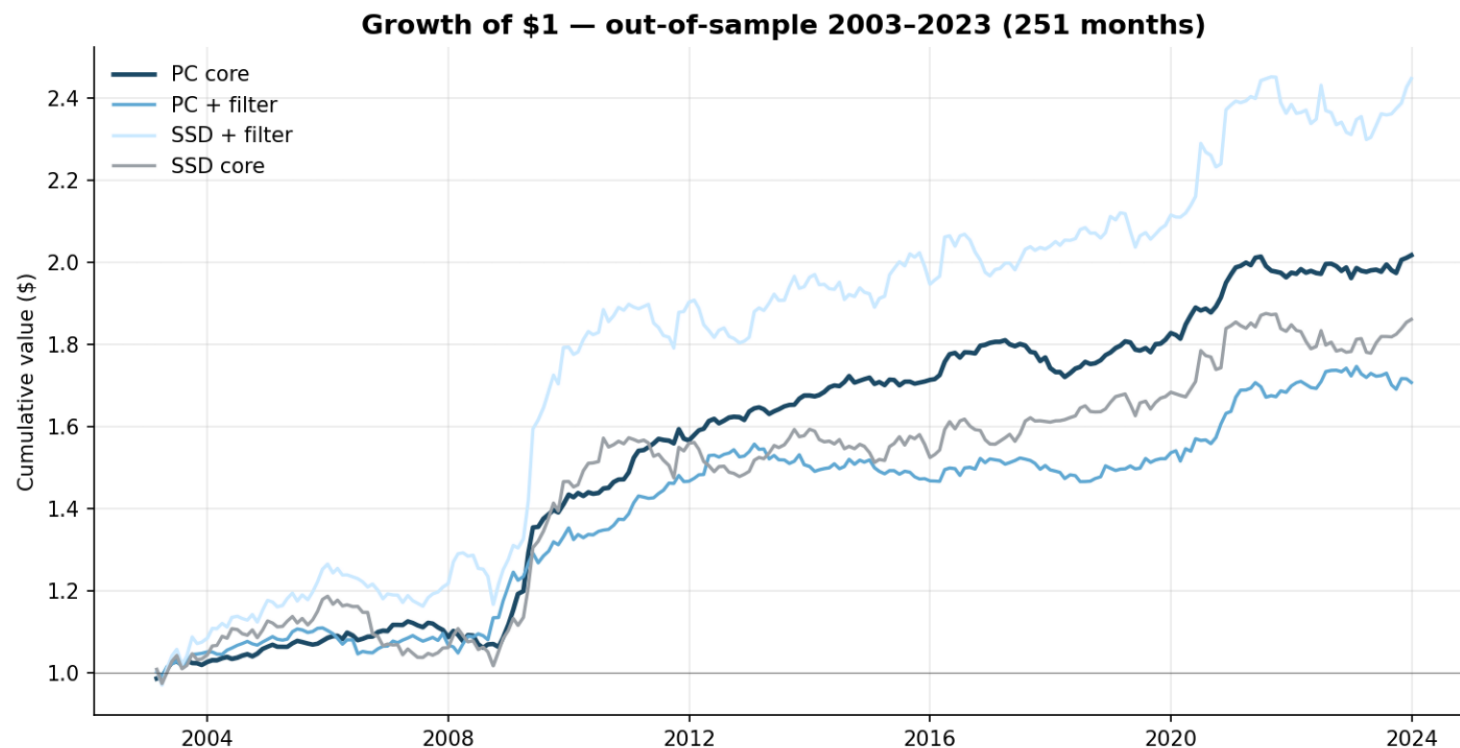
Matched the paper — across all four variants



PC core: Sharpe 1.028 vs the paper's 1.01 — inside the ± 0.15 tolerance band. Every variant lands where the paper says it should.

RESULTS

Growth of \$1 — 251 months out-of-sample



PC core (dark) is the smoothest compounder. SSD+filter ends higher in absolute terms but is far choppier — which is exactly why its risk-adjusted Sharpe is lower.

RESULTS

Full scorecard

Strategy	Ann. ret	Ann. vol	Sharpe	Max DD	vs paper
PC core	3.41%	3.32%	1.028	-5.7%	1.01 ✓
PC + filter	2.59%	3.48%	0.752	-5.9%	0.80 ✓
SSD + filter	4.37%	6.11%	0.731	-9.7%	0.75 ✓
SSD core (baseline)	3.01%	5.28%	0.589	-14.3%	0.88 ✗

**PC roughly halves both volatility AND drawdown vs SSD —
the market-neutral, idiosyncratic pairs are simply lower-risk.**

Why it works — diagnosed, not assumed

I ran a P&L attribution on the baseline first and found a bimodal pattern:

11.4% of trades

cleanly revert: +471 bps each

→ *ALL of the profit*

88.4% of trades

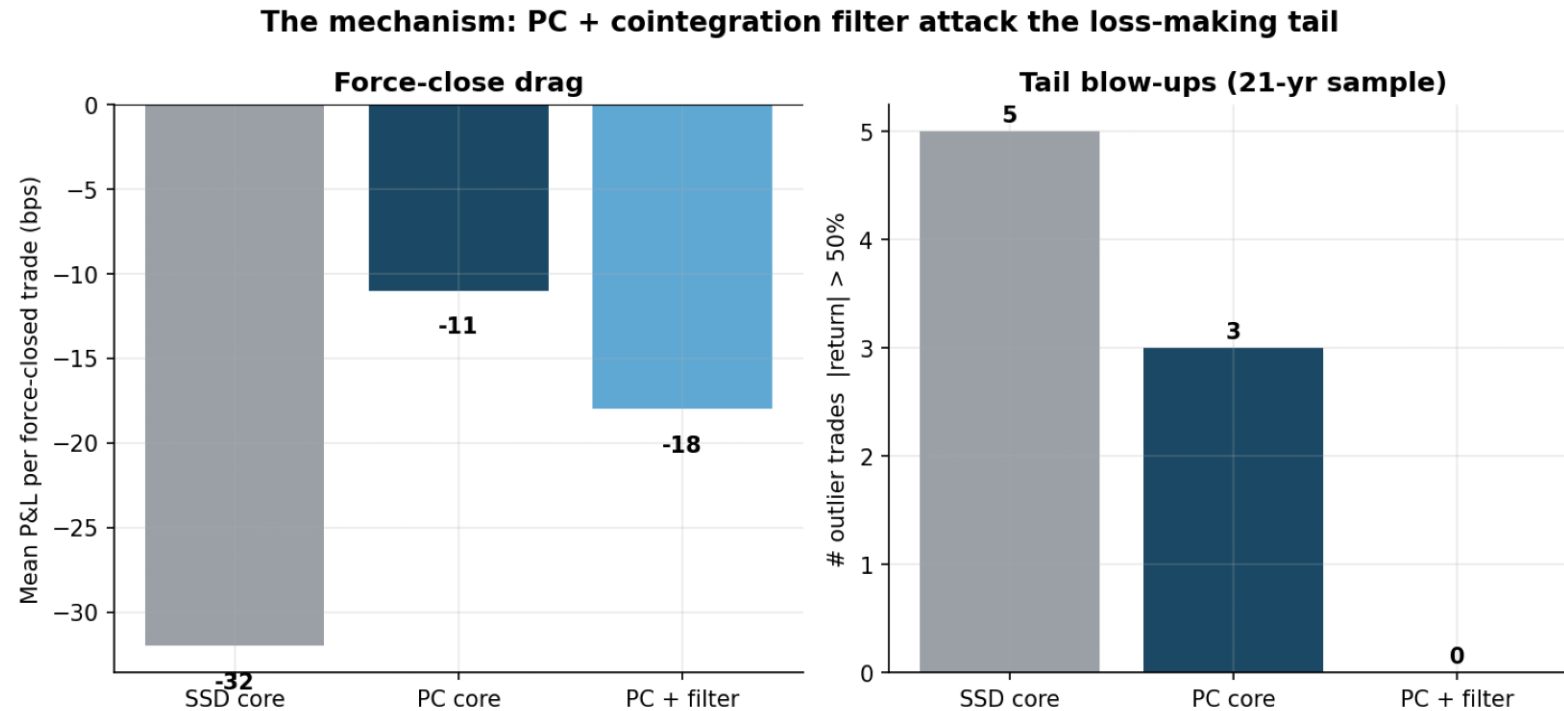
force-closed: -32 bps each

→ *a constant drag*

Thesis

The job isn't "find a higher Sharpe." It's KILL THE FORCE-CLOSE DRAG — only trade pairs that actually revert.

Mechanism — prediction confirmed



PC distance cut per-trade force-close drag by 65%; adding the cointegration filter eliminated every tail blow-up (0 trades with $|return| > 50\%$ in 21 years, vs 5 for the baseline).

Built like production research code

- Modular package — distances · clustering · spread · cointegration · backtest · performance, each independently tested.
- 18 synthetic unit tests — validate every component on data with a KNOWN answer before touching real CRSP.
- Look-ahead protection baked into the z-score and the rolling formation/trading split.
- Invariant checks — the extended engine reproduces the Phase-1 baseline Sharpe to 4 decimal places (no regression).
- Reproducible — config-driven single source of truth; hyperparameters frozen pre-results.

What this demonstrates

- Independent replication of a 2025 research paper — read it, rebuilt it, matched the headline number on self-sourced data.
- Quant intuition — diagnosed WHY the strategy makes money, then engineered the fix and confirmed the prediction.
- Discipline — survivorship-bias-free data, out-of-sample design, frozen hyperparameters, unit-tested code.

Roadmap

Factor-beta clustering (my own extension) → robustness suite (hierarchical clustering, robust hedge ratios, stop-losses) → live Alpaca paper-trade forward test.

Thank you

Clustering-Based Pairs Trading

Sharpe 1.028 · 21 years out-of-sample · built from the data up

Happy to walk through the code, the backtest engine, or the attribution analysis.